Lorentz Transformations from Projective Temporal Ordering

LC-O1 of the Lorentz–Capacity Sub-Programme

Jérôme Beau

Independent Researcher

jerome.beau@cosmochrony.org

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Abstract

We close LC-O1, the reconstruction of effective Lorentz transformations between inertial observers from projective temporal ordering data. The Lorentz-capacity construction of [5, 6] establishes that the projective temporal capacity satisfies

$$F^\tau = \sqrt{1-\beta^2} = \frac{1}{\gamma},$$

where β is the normalised spatial capacity load and γ is the Lorentz factor, closing time dilation and the light cone. The remaining open problem is the reconstruction of the full observer transformation, including length contraction. We prove that an inertial observer corresponds to a unit timelike splitting of the completed effective co-metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ established in Q5b–Q11 [2, 7]. Given two such splittings with relative capacity load β, the unique proper orthochronous isometry of the reconstructed flat interval that maps one temporal axis to the other is the standard Lorentz boost

$$\Lambda(\beta) = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix}.$$

Length contraction follows as the reciprocal spatial projection onto the simultaneity hyperplane of the moving splitting:

$$\ell = \frac{\ell_0}{\gamma}.$$

The Born–Infeld capacity sphere parameterises admissible capacity splittings and is not a Lorentzian structure; the Lorentz group acts as the isometry group of the effective metric η, not of the BI sphere. Special-relativistic observer transformations are thereby reconstructed from the same capacity data that produced projective time dilation.

Keywords. Lorentz transformation; length contraction; inertial splitting; projective temporal ordering; Born–Infeld capacity sphere; effective Lorentzian metric; Cosmochrony.

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1 Introduction

The Lorentz-capacity sub-programme derives relativistic kinematics from the bounded Born–Infeld capacity budget of admissible projection on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$. Paper [5] identifies the projective temporal capacity $F^\tau = \sqrt{1 - B^2}$ as the Lorentz factor residual, and [6] proves that the temporal residual map

$$P_\tau^{Q5b}\Sigma_n^{\mathrm{tot}} = \sqrt{1 - B_n^2}$$

is the unique positive function compatible with the Lorentzian splitting of Q5b–Q11, the Born–Infeld unit capacity sphere, and the Weil filling law. This closes time dilation ($d\tau/dn = 1/\gamma$) and the light cone ($B = 1 \Rightarrow F^\tau = 0$).

The open problem LC-O1 is stated in Section 7.2 of [6]: length contraction requires the reconstruction of effective Lorentz transformations between inertial observers from projective temporal ordering data.

The resolution rests on a strict separation of two structures. The Born–Infeld capacity sphere

$$(F^\tau)^2 + \beta^2 = 1$$

is a Euclidean capacity relation. It parameterises admissible temporal splittings and determines β and $1/\gamma$, but it is not the Minkowski interval. The Lorentzian metric is supplied by the homogeneous Q5b–Q11 closure:

$$g^{\mu\nu} = 2\eta^{\mu\nu}.$$

Lorentz transformations are the isometries of η, not symmetries of the BI sphere. The BI sphere determines which splittings are admissible; the Lorentz group determines how to pass between them.

Section 2 defines inertial observers as unit timelike splittings of the effective metric and identifies the rapidity with the projective capacity load. Section 3 proves that the unique interval-preserving isometry mapping one splitting to another is the Lorentz boost. Section 4 derives length contraction. Section 5 states the connection to projective temporal ordering. Sections 6 and 7 summarise the logical structure.

2 Inertial splittings and capacity load

Let M_{eff} be the homogeneous effective spacetime reconstructed from the admissibility operator. The Q5b–Q11 closure gives the completed effective cometric

$$g^{\mu\nu} = 2\eta^{\mu\nu}, \qquad \eta = \mathrm{diag}(-1, +1, +1, +1),$$

with no free parameter [2, 7]. The corresponding metric is $g_{\mu\nu} = \frac{1}{2}\eta_{\mu\nu}$. The isometry group of g is $\mathrm{SO}(3, 1)$, the same as that of η.

Definition 2.1 (Inertial splitting)**.** An inertial observer is a future-directed unit timelike vector $u \in T_x M_{\text{eff}}$ satisfying $\eta(u,u) = -1$, together with its orthogonal simultaneity hyperplane

$$\Sigma_u := \big\{ X \in T_x M_{\text{eff}} \mid \eta(X,u) = 0 \big\}.$$

Remark 2.2 (Capacity sphere versus Lorentzian structure). The Born–Infeld capacity sphere $(F^\tau)^2 + \beta^2 = 1$ is a Euclidean norm constraint on the capacity flow vector, not a Lorentzian structure. It provides the values of β and $1/\gamma$ from projective data, but it does not determine the Lorentzian interval. The Lorentzian interval is supplied by η from the Q5b–Q11 geometric closure [2, 7]. The Lorentz group acts as isometries of η; it does not act as isometries of the BI sphere.

The rest observer corresponds to the splitting $u_0 = \partial_\tau$. An observer with normalised spatial capacity load β in the x-direction has splitting

$$u_\beta = \gamma(\partial_\tau + \beta\,\partial_x), \qquad \gamma := \frac{1}{\sqrt{1-\beta^2}}.$$

One verifies directly that u_β is unit timelike: $\eta(u_\beta, u_\beta) = -\gamma^2 + \gamma^2\beta^2 = -1$.

The parameter β is not introduced independently. By the Lorentz-capacity result of [5, 6], the projective temporal capacity satisfies

$$F^\tau = \sqrt{1-\beta^2} = \frac{1}{\gamma},$$

so the rapidity of the inertial splitting is determined by the projective temporal capacity already computed in [6].

3 Boost reconstruction

Let ξ be the rapidity defined by

$$\beta = \tanh\xi, \qquad \gamma = \cosh\xi, \qquad \gamma\beta = \sinh\xi.$$

Then $u_\beta = \cosh\xi\,\partial_\tau + \sinh\xi\,\partial_x$.

Lemma 3.1 (Uniqueness of the inertial boost)**.** *There is a unique proper orthochronous linear isometry of $(T_x M_{\text{eff}}, \eta)$ that maps $u_0 = \partial_\tau$ to u_β and acts as the identity on the transverse directions ∂_y, ∂_z. It is the Lorentz boost of rapidity ξ, acting on the (τ, x)-plane by*

$$\begin{pmatrix} \partial_{\tau'} \\ \partial_{x'} \end{pmatrix} = \begin{pmatrix} \cosh\xi & \sinh\xi \\ \sinh\xi & \cosh\xi \end{pmatrix} \begin{pmatrix} \partial_\tau \\ \partial_x \end{pmatrix}.$$

Proof. A proper orthochronous linear isometry of η preserving ∂_y and ∂_z acts only on the (τ, x)-plane. Preservation of the restriction $-d\tau^2 + dx^2$ requires a hyperbolic rotation: the isometry is of the form displayed above for some ξ. The condition $\partial_{\tau'} = u_\beta$ uniquely fixes $\cosh\xi = \gamma$ and $\sinh\xi = \gamma\beta$, hence $\xi = \operatorname{arctanh}\beta$. □

In coordinate form the boost reads

$$\tau' = \gamma(\tau - \beta x), \qquad x' = \gamma(x - \beta\tau), \qquad y' = y, \qquad z' = z.$$

Theorem 3.2 (Lorentz transformations from projective temporal ordering)**.** *Given a projective temporal capacity load β determined by [5, 6], the unique proper orthochronous transformation between the corresponding inertial splittings is the Lorentz boost*

$$\tau' = \gamma(\tau - \beta x), \qquad x' = \gamma(x - \beta\tau), \qquad y' = y, \qquad z' = z,$$

with $\gamma = (1 - \beta^2)^{-1/2}$.

Proof. The projective temporal capacity result of [5, 6] gives $F^\tau = (1 - \beta^2)^{1/2} = 1/\gamma$, fixing β and hence ξ. The Q5b–Q11 closure gives the effective co-metric $g^{\mu\nu} = 2\eta^{\mu\nu}$. Since the factor 2 is a global conformal rescaling, the isometry group of g is $\mathrm{SO}(3,1)$, the isometry group of η. By Lemma 3.1, the unique proper orthochronous isometry of η mapping ∂_τ to u_β and fixing the transverse directions is the displayed Lorentz boost. □

4 Length contraction

Corollary 4.1 (Length contraction)**.** *A rod of proper length ℓ_0 at rest in the unprimed splitting has length $\ell = \ell_0/\gamma$ when measured on the simultaneity hyperplane of the primed splitting.*

Proof. Let the rod endpoints in the rest frame be $A = (\tau_A, x_A)$ and $B = (\tau_B, x_B)$ with $\Delta x = x_B - x_A = \ell_0$. The moving observer measures length at equal primed time, $\Delta\tau' = 0$. The boost gives $\Delta\tau' = \gamma(\Delta\tau - \beta\Delta x)$, so simultaneous in the primed frame means $\Delta\tau = \beta\Delta x$. The primed spatial separation is then

$$\Delta x' = \gamma(\Delta x - \beta\Delta\tau) = \gamma(1 - \beta^2)\Delta x = \frac{\Delta x}{\gamma} = \frac{\ell_0}{\gamma}.$$

□

5 Connection to projective temporal ordering

The projective temporal ordering observable [4] supplies the scalar clock factor

$$\frac{d\tau}{dn} = F^{\tau} = \frac{1}{\gamma}.$$

LC-O1 shows that this factor is not an isolated time-dilation effect. Once the effective Lorentzian interval is reconstructed, the same capacity load β determines the rapidity $\xi = \operatorname{arctanh}\beta$ and hence the complete inertial transformation.

Projective temporal ordering determines the clock axis of the observer. Q5b–Q11 determine the Lorentzian interval on which this axis lives. The boost is the unique interval-preserving change of admissible inertial splitting.

6 Discussion

The construction separates three layers, each with a distinct source.

Layer 1: Born–Infeld capacity sphere. The bounded BI admissibility budget of [3] gives the unit capacity relation

$$(F^{\tau})^2 + \beta^2 = 1.$$

This is a Euclidean norm constraint on the capacity flow vector. It is not the Minkowski interval. It provides β and $1/\gamma$ from projective data.

Layer 2: Effective Lorentzian co-metric. The homogeneous geometric closure of Q5b–Q11 [2, 7] gives the effective co-metric

$$g^{\mu\nu} = 2\eta^{\mu\nu}.$$

This supplies the interval whose isometry group is $\mathrm{SO}(3,1)$. The global factor 2 is physically irrelevant for the isometry group.

Layer 3: Inertial observer and boost. An inertial observer is a unit timelike splitting of η. The rapidity of a splitting is fixed by the projective capacity load β from Layer 1. The Lorentz boost is the unique isometry of η (Layer 2) that maps one admissible splitting to another at capacity load β.

Length contraction is a consequence of this boost, not an independent capacity law. It follows from the difference in simultaneity hyperplanes between two inertial splittings, given the interval structure of Layer 2.

Remark 6.1 (Logical status). All inputs are previously closed: time dilation and $F^{\tau} = 1/\gamma$ in [5, 6]; the effective co-metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ in [2, 7]; the BI admissibility budget in [3, 1]. LC-O1 adds only the identification of inertial

observers with unit timelike splittings and the application of the standard uniqueness of isometries of η in each rapidity class. No new spectral or projective hypothesis is introduced.

7 Conclusion

LC-O1 is closed. Projective temporal ordering data determine the capacity load β and the residual $F^\tau = 1/\gamma$. The Q5b–Q11 closure supplies the flat effective Lorentzian interval with isometry group $\mathrm{SO}(3,1)$. An inertial observer is a unit timelike splitting of this interval; changing observer is changing the splitting while preserving the interval. The unique proper orthochronous isometry at capacity load β is the Lorentz boost:

$$\tau' = \gamma(\tau - \beta x), \qquad x' = \gamma(x - \beta\tau).$$

Consequently, a rest length ℓ_0 is measured as $\ell = \ell_0/\gamma$.

The full Lorentz transformation, including length contraction, is thereby reconstructed from projective temporal ordering data and the effective metric normalisation. The Born–Infeld capacity sphere provides β and γ; it is not the Minkowski interval and does not carry the Lorentz group as an isometry.

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